

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2
Hello World
Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

// You are using Java

```
import java.util.*;
```

```
class main
```

```
{
```

```
    public static void main(String[] args)
```

```
{
```

```
    Scanner sc = new Scanner(System.in);
```

```
    int n= sc.nextInt();
```

```
    TreeMap <Character,Integer> freqMap = new TreeMap<>();
```

```
    for(int i=0;i<=n;i++)
```

```
{
```

```
String s = sc.nextLine();
for(char ch : s.toCharArray())
{
    if(Character.isLetter(ch))
    {
        freqMap.put(ch,freqMap.getOrDefault(ch,0)+1);
    }
}

System.out.println("Character Frequency: ");
for(Map.Entry<Character,Integer>entry: freqMap.entrySet())
{
    System.out.println(entry.getKey()+" : "+entry.getValue());
}
}
```

Status : Correct

Marks : 10/10